BioTrack / AuthRos Platform Overview

Comprehensive Project Outline & Technical Status Report | August 2026

1. Executive Summary

BioTrack (AuthRos) is a centralized biomedical research, clinical trial tracking, and public crowdfunding platform. It connects Principal Investigators (PIs), researchers, healthcare institutions, and public supporters into a unified ecosystem.

2. Technical Architecture & Stack

Layer	Technologies Implemented
Frontend UI	HTML5, Tailwind CSS, FontAwesome 6, Vanilla JS (No heavy framework lock-in)
Backend API	Node.js, Express.js REST API (`server.js`)
Database	SQLite (`Database/biotrack.db`) with `init-db.js` re-seeding engine
Version Control	Git & GitHub Repository: `https://github.com/MrBlackJags/AuthRos.git`

3. Implemented Modules & Features

• **Global Research Archive (246+ Papers):** Spanning 1928–2026 across 12+ lead countries (India ■■, USA ■■, UK ■■, Germany ■■, Japan ■■, Switzerland ■■, Sweden ■■, France ■■, etc.) and official publishers (Nature, Science, NEJM, Lancet, Cell, PNAS).

• **Interactive View Layout Switcher:** 4 distinct display modes (Card View, List View, Concise Table View for high-density browsing, Brief Abstract Reader View).

• **Multi-Language Translation Engine:** Live switcher supporting English, Spanish, French, German, Japanese, and Hindi.

• **Research Domains Hub (`public/research-domains.html`):** Categorizes studies under 6 major disciplines (Genomic Medicine, Oncology, Neuroscience, Synthetic Biology, Cardiology, Cell Therapy).

• **Personal Saved Research Library (`public/saved-research.html`):** User bookmarks & liked studies library with 1-click APA / BibTeX bibliography exports.

• **Dark / Light Mode Theme System:** Persistent theme toggle with inline FOUC prevention across all pages.

• **Crowdfunding & Patient Cohorts:** REST API integrations for public donations and hospital de-identified cohort record submissions.

4. Strategic Next Steps (User Masterplan)

1. **Phase 1 (Upcoming UI/UX Redesign):** Complete modernization of the user interface. Replacing default Tailwind presets with custom glassmorphic panels, elevated typography, fluid micro-interactions, and high-end design tokens.

1. **Phase 2 (Research Domain Deep Dive):** Advanced tooling for research paper annotations, interactive dataset viewing, and peer review submissions.

1. **Phase 3 (Clinical & Crowdfunding Domain Overhaul):** Refining and polishing the remaining 2 core domains (Hospital Patient Cohort Portal & Public Crowdfunding Portal).